Fair Data Management in a Cloud Based Healthcare Platform

D4F Assignment

Use case: Fair Data Management in a Cloud Based Healthcare Platform

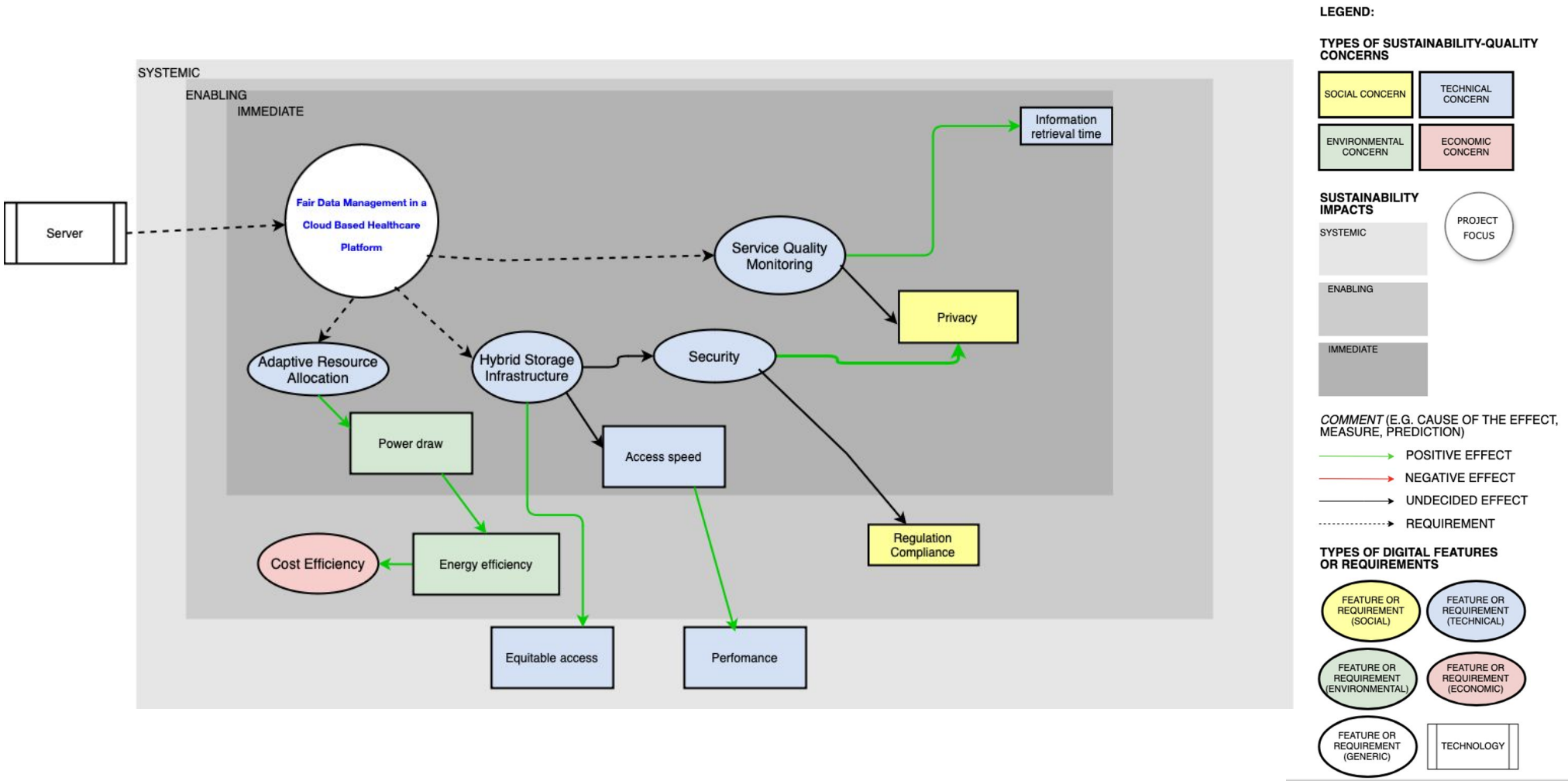
Persona: Tom has to visit the hospital due to a illness and while doing so has to provide his data. one this data is provided, it is stored in a central database that can be queried by the medical institution for any subsequent visits. But this time Tom has visited the hospital but the data is taking very long to retrieve and is costing valuable time as Tom is very sick.

Context: A cloud based healthcare platform stores patient data (eg. medical records, test results) across multiple data centers. The platform serves hospitals, clinics and individual patients.

Fairness Concern: Ensure equitable access to patient data for all stakeholders (e.g., hospitals, clinics, patients) while maintaining data security, privacy, and cost efficiency.

Challenge: Some data is "hot" (frequently accessed, e.g., recent patient records), while other data is "cold" (rarely accessed, e.g., archived records). The system must balance storage costs, access speed, and energy consumption while ensuring fairness in data availability and accessibility.

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

To ensure equitable healthcare data access for all stakeholders—including rural clinics and underserved populations—the system combines tiered infrastructure, adaptive resource allocation, and energy-aware

Tiered Cloud-Edge Infrastructure strategically places SSDs at edge nodes near rural clinics for critical data (<2s latency), while using centralized HDDs for cost-efficient archival storage. Frequently accessed records (e.g., active patient charts) are cached at the edge to overcome connectivity gaps, improving availability and reducing geographic disparities.

Adaptive Resource Allocation dynamically prioritizes requests based on stakeholder role (hospitals > clinics > patients) and data criticality (e.g., ER admissions over routine checks). Real-time monitoring detects and mitigates latency imbalances, directly supporting healthcare equity.

Energy-Aware Scheduling routes queries to renewable-powered data centers during peak loads, though this may slightly delay non-urgent requests which are a deliberate trade-off for environmental fairness.

Outcomes: Balanced urban-rural care disparities and Compliance with privacy and carbon accountability directives.

Key Trade-off: Energy efficiency may marginally impact non-critical query latency (context-dependent)

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION OF METRIC/KPI
ENERGY EFFICIENCY	Minimizes energy consumption per data operation	ISO/IEC 30134-2:2016		X			Joules per query
COST EFFICIENCY	Balances storage/access costs across stakeholders	ISO/IEC 19770-1:2017			X		Storage cost per GB
EQUITABLE ACCESS	Ensures equal data access for all stakeholders (hospitals, clinics, patients)	ISO/IEC 29100:2011				X	% requests fulfilled within SLA
PERFORMANCE	Guarantees low-latency data access	ISO/IEC 25023:2016	X				Average retrieval time (ms)
PRIVACY	Protects sensitive patient data per regulations (GDPR/HIPAA)	ISO/IEC 27018:2019				X	% data encrypted
DATA FRESHNESS	Ensures timely data updates	ISO 8000-2:2022	X				Time to update records(ms)

Definition of Fairness: SQ model (cont.)

The SQ Model maps fairness concerns to sustainability dimensions, ensuring that the system is environmentally friendly, economically viable, and socially equitable.

- Energy Efficiency: Reduces energy consumption for "cold" data storage. This is environmental because it directly affects the power draw of the entire system.

- Cost Efficiency: Balances storage costs across stakeholders. This is economic as it impacts the financial feasibility of the system in the long run.

- Equitable Access: Ensures all stakeholders have equal access to data. This is social as it concerns the right of the stakeholders to all have equal and just access to the data.

- Performance: Measures the latency rate of data passthrough and hence gives a fair indication on operating efficiency. This is technical as it directly relates the systems functional capability.

- Privacy: Checks on data storage as per contemporary regulations and standards. This is social as it concerns the rights of the stakeholders to have their personal medical data not be divulged to third parties without explicit consent.

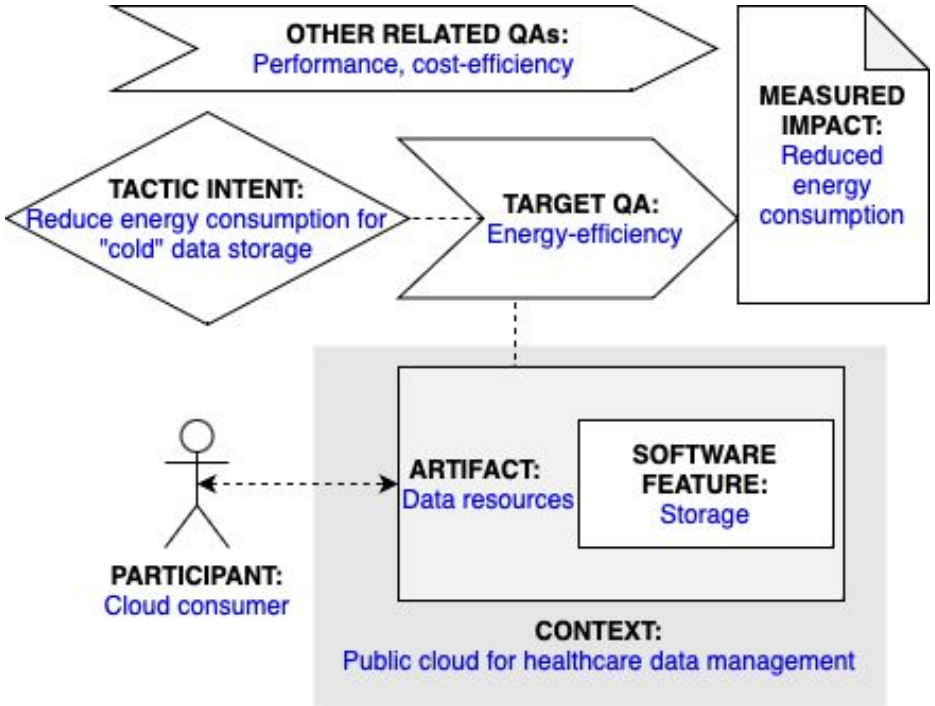
- Data freshness: In very simple terms, it ensures that the data accessible is always up to date and hence termed as “fresh”. This is technical as it concerns the quality of the output of the system and an lack of freshness can make the whole system pointless with old data.

Definition of KPI*

Percentage of Data Accessible within SLA					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Ensure equitable access to data for all stakeholders	All stakeholders should have access to data within the agreed SLA (e.g., 99.9% uptime)	Percentage of data accessible within SLA = (Number of successful data accesses / Total data access requests) * 10		Number of successful data accesses Total data access requests to the server Latency per request over a time scale Response times at peak and off peak load hours	Count of successful data accesses Count of total data access requests Average response time over a number of requests
		Target	**Action(s)**		
		99.9% of data accessible within SLA Higher overall KPI score	Implement load balancing and failover mechanisms to ensure high availability		

SLA stands for "Service Level Agreement"

Architectural Tactic(s) for Fairness*

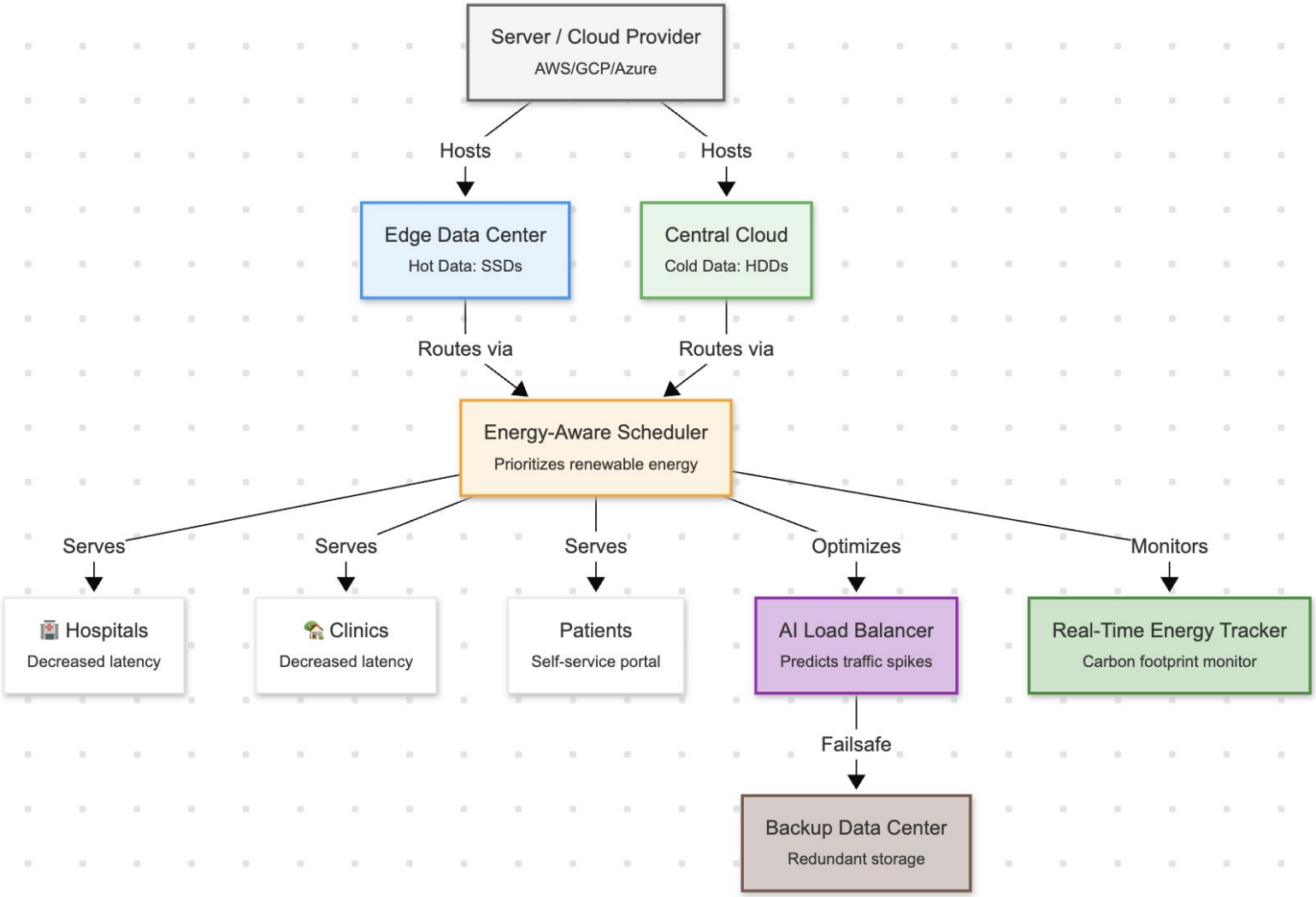


TITLE	Compress infrequently accessed data
DESCRIPTION	Patient data that is rarely accessed (cold data) is dynamically identified and compressed to optimize storage costs and reduce energy consumption. A dynamic threshold differentiates between "hot" (frequently accessed) and "cold" data. If data falls below the access frequency threshold, it is compressed. An adaptive, yet lightweight, monitoring mechanism observes data access patterns to adjust thresholds as needed. Although this approach introduces some CPU overhead for (de)compression, in our healthcare context this trade-off is acceptable since our primary focus is on maintaining fairness: ensuring equitable data availability to all stakeholders (hospitals, clinics, and patients) without excessive cost or energy use.
PARTICIPANT	Cloud Consumer (e.g., the healthcare platform operator)
ARTIFACT	Data resources (storage systems and patient data repositories)
CONTEXT	Public Cloud environment for healthcare data management
SOFTWARE FEATURE	Storage
TACTIC INTENT	Reduces energy consumption for “cold” data storage
TARGET QA	Cost-efficiency and energy-efficiency
OTHER RELATED QA’S	Performance, …
MEASURED IMPACT	The percentage of cold data remaining accessible within defined SLAs, equitable data availability across stakeholder groups, reduced storage costs and energy consumption per GB of compressed data.
REFERENCES	* S. Vos, P. Lago, R. Verdecchia and I. Heitlager, "Architectural Tactics to Optimize Software for Energy Efficiency in the Public Cloud," *2022 International Conference on ICT for Sustainability (ICT4S)*, Plovdiv, Bulgaria, 2022, pp. 77-87, doi: 10.1109/ICT4S55073.2022.00019. - Len Bass, Paul Clements, and Rick Kazman (2021) Software Architecture in Practice, 4th Edition. Addison-Wesley Professional, p.219-220

Fairness perspective

	DESCRIPTION
DESIRED QUALITY	Fairness in healthcare Data Management: A system where all stakeholders (hospitals, clinics, patients) receive equitable access to data resources while balancing cost efficiency, energy sustainability, and regulatory compliance, as measured by ISO-standardized KPIs.
CONCERNS	*Equitable access:* Ensure rural clinics have comparable latency to urban hospitals (<5s difference) and Prevent bias in prioritization (e.g., no preferential treatment for high-revenue hospitals). *Cost efficiency/ transparency:* Clear cost allocation per stakeholder group (e.g., storage costs/GB/month). *Energy efficiency:* Reduce the data retrieval costs while also maintaining high access rate. *Performance:* The satisfaction of the systems users and the overall average of all metrics which gives us an idea of how well the system works. *Privacy & Security:* Maintaining adherence to GDPR and other international standards.
ARCHITECTURAL TACTIC(S)	Tiered storage, energy-aware scheduling, **compress infrequently accessed data.**
POSSIBLE KPI(S)	Percentage of data accessible within SLA, energy consumption per query.
PROBLEMS AND PITFALLS	Balancing performance vs. energy efficiency, ensuring compliance with privacy regulations.

Fairness view: Deployment View for Fair Data Management



Fairness view (cont.)

- The fairness view illustrates how the system ensures equitable access to data while balancing cost, performance, and energy efficiency.

- The energy-aware scheduler optimizes data retrieval processes, ensuring that "hot" data is accessed quickly while "cold" data is stored in an energy-efficient manner.

- The role of the AI load balancer is to monitor data flow and predict future traffic spikes. Just like a load balancer in an electrical circuit, it prevents huge spikes of data access at busy hours of the day for example and maintains fairness by making sure the system is capable of such spikes.

- The edge data centre hosts hot data storage mediums such as SSDs which are better for faster more frequent queries which is common with recently acquired data.